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Qualification Pack

MERN Stack Developer

QP Code: ASP/ITS/Q0501

Version: 1.0

NSQF Level: 5

Additional Skill Acquisition Programme Kerala || Additional Skill Acquisition Programme Kerala, 3rd
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Qualification Pack

Contents

ASP/ITS/Q0501: MERN Stack Developer	3
<i>Brief Job Description</i>	3
Applicable National Occupational Standards (NOS)	3
<i>Compulsory NOS</i>	3
<i>Qualification Pack (QP) Parameters</i>	3
ASP/ITS/N0501: Introduction to MERN	5
ASP/ITS/N0502: Javascript and Typescript Fundamentals	8
ASP/ITS/N0503: Building Web applications with React and Next.js	11
ASP/ITS/N0504: Building with Database: MongoDB for developers	14
ASP/ITS/N0505: NodeJS for Backend development	17
ASP/ITS/N0506: On-the-job training	22
ASP/ITS/N0507: PROJECT ON MERNSTACK DEVELOPER COURSE	25
DGT/VSQ/N0102: Employability Skills (60 Hours)	28
Assessment Guidelines and Weightage	35
<i>Assessment Guidelines</i>	35
<i>Assessment Weightage</i>	37
Acronyms	39
Glossary	40

Qualification Pack

ASP/ITS/Q0501: MERN Stack Developer

Brief Job Description

This comprehensive certificate program equips the students to build modern, interactive web applications with frameworks like ReactJS and Next.js, while mastering data management with databases like MySQL and MongoDB, all powered by server-side development with Node.js and modern development practices.

Personal Attributes

Attention to Detail, Analytical Thinking, Problem-Solving Skills, Creativity, Adaptability, Persistence, Team Collaboration, Time Management, Communication Skills

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [ASP/ITS/N0501: Introduction to MERN](#)
2. [ASP/ITS/N0502: Javascript and Typescript Fundamentals](#)
3. [ASP/ITS/N0503: Building Web applications with React and Next.js](#)
4. [ASP/ITS/N0504: Building with Database: MongoDB for developers](#)
5. [ASP/ITS/N0505: NodeJS for Backend development](#)
6. [ASP/ITS/N0506: On-the-job training](#)
7. [ASP/ITS/N0507: PROJECT ON MERNSTACK DEVELOPER COURSE](#)
8. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Information Technology Sector
Sub-Sector	BIOINFORMATICS
Occupation	MERN Stack Developer
Country	India
NSQF Level	5

Qualification Pack

Credits	18
Aligned to NCO/ISCO/ISIC Code	2512.0204
Minimum Educational Qualification & Experience	Completed 2nd year of UG (UG Diploma) (any Engineering degree, BCA,BSC – CS/IT/Circuit branches any Engineering degree, BCA,BSC – CS/IT/Circuit branches) with NA of experience Basic programming knowledge is recommended OR Completed 2nd year diploma after 12th (any Engineering degree, BCA,BSC – CS/IT/Circuit branches) OR Completed 3 year diploma after 10th (any Engineering degree, BCA,BSC – CS/IT/Circuit branches) with 1 Year of experience OR Completed 1st year of UG (UG Certificate) (any Engineering degree, BCA,BSC – CS/IT/Circuit branches) with 1 Year of experience relevant OR 12th Class (pass) with 2 Years of experience relevant
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	18/02/2028
NSQC Approval Date	18/02/2025
Version	1.0
Reference code on NQR	QG-05-IT-03518-2025-V1-ASAP
NQR Version	1

Qualification Pack

ASP/ITS/N0501: Introduction to MERN

Description

This NOS equips students with the knowledge and practical skills needed to build structured and visually appealing web pages using HTML and CSS. It covers web development basics, including the client-server model, HTTP/HTTPS protocols, and database interactions. Students will gain hands-on experience with HTML elements, forms, and semantic tags, as well as styling techniques using CSS for responsive design. Additionally, they will explore advanced CSS techniques like positioning, pseudo-classes, and pre-processors to enhance their web pages. This module serves as the foundational step toward full-stack development using the MERN stack.

Scope

The scope covers the following :

- This covers the foundational concepts and technologies required for MERN Stack development.
- It includes an introduction to web development, HTML, and CSS, ensuring students gain a strong understanding of front-end development.
- The scope includes learning about web page structures, styling techniques, responsive design, and fundamental client-server communication principles.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Demonstrate theoretical knowledge of web development fundamentals ,HTML functionalities, and CSS styling techniques.
- PC2.** Write clean, well-structured, and commented HTML and CSS code to create basic web pages and apply styling.
- PC3.**
- Build responsive layouts that adapt to different screen sizes.
 - Implement form functionalities and handle user input data.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	10	10	-	-
PC1. Demonstrate theoretical knowledge of web development fundamentals ,HTML functionalities, and CSS styling techniques.	3	4	-	-
PC2. Write clean, well-structured, and commented HTML and CSS code to create basic web pages and apply styling.	3	3	-	-
PC3. - Build responsive layouts that adapt to different screen sizes. - Implement form functionalities and handle user input data.	4	3	-	-
NOS Total	10	10	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	ASP/ITS/N0501
NOS Name	Introduction to MERN
Sector	Information Technology Sector
Sub-Sector	
Occupation	MERN Stack Developer
NSQF Level	5
Credits	1
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Qualification Pack

ASP/ITS/N0502: Javascript and Typescript Fundamentals

Description

This NOS provides learners with a comprehensive understanding of JavaScript and TypeScript, enabling them to build dynamic and interactive web applications. It covers core programming concepts, including variables, functions, objects, classes, and asynchronous operations using promises and async/await. Students will gain hands-on experience with DOM manipulation, browser APIs, and modular JavaScript. The module also introduces TypeScript to enhance code maintainability, focusing on type annotations, interfaces, function overloading, and generics. Furthermore, students will develop expertise in implementing and optimizing data structures such as arrays, linked lists, stacks, queues, trees, and graphs. The NOS ensures learners can analyze algorithmic complexity, select appropriate data structures, and optimize code for efficiency, preparing them for real-world software development challenges in the MERN stack ecosystem.

Scope

The scope covers the following :

- This NOS covers the essential programming skills required for MERN Stack development, focusing on JavaScript, TypeScript, and data structures.
- It includes fundamental and advanced concepts of JavaScript, asynchronous programming, DOM manipulation,
- TypeScript type annotations, generics, and object-oriented programming. Additionally, it encompasses the implementation of key data structures and their applications in problem-solving and performance optimization.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Demonstrate a solid understanding of JavaScript fundamentals (syntax, variables, data types, operators, control flow).
- PC2.** Write clean and efficient JavaScript code to perform basic operations, manipulate data structures (arrays, objects), and interact with the DOM.
- PC3.** Handle asynchronous tasks effectively using promises, async/await, and callbacks to create responsive web applications.
- PC4.** Apply type annotations in TypeScript code to improve code clarity, reduce errors, and enhance code maintainability.
- PC5.**
 - Effectively utilize TypeScript features like interfaces, functions, classes,
 - and modules to organize and structure code.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	10	10	-	-
PC1. Demonstrate a solid understanding of JavaScript fundamentals (syntax, variables, data types, operators, control flow).	2	2	-	-
PC2. Write clean and efficient JavaScript code to perform basic operations, manipulate data structures (arrays, objects), and interact with the DOM.	2	2	-	-
PC3. Handle asynchronous tasks effectively using promises, async/await, and callbacks to create responsive web applications.	2	2	-	-
PC4. Apply type annotations in TypeScript code to improve code clarity, reduce errors, and enhance code maintainability.	2	2	-	-
PC5. - Effectively utilize TypeScript features like interfaces, functions, classes, - and modules to organize and structure code.	2	2	-	-
NOS Total	10	10	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	ASP/ITS/N0502
NOS Name	Javascript and Typescript Fundamentals
Sector	Information Technology Sector
Sub-Sector	
Occupation	MERN Stack Developer
NSQF Level	5
Credits	3
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

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ASP/ITS/N0503: Building Web applications with React and Next.js

Description

This NOS provides learners with the expertise to develop dynamic and interactive web applications using ReactJS and Next.js. Students will gain proficiency in creating reusable components, managing application state with Redux/Zustand, and implementing efficient client-side routing using React Router. The training includes hands-on experience with Next.js, covering server-side rendering (SSR), static site generation (SSG), and API integrations for optimized web performance. Learners will also be introduced to TypeScript with React for enhanced code maintainability. Additionally, the NOS covers unit testing fundamentals using Jest, equipping students with the ability to write, execute, and analyze test cases for ensuring code reliability. Through practical exercises, students will develop scalable and maintainable applications while following industry best practices in modern JavaScript development.

Scope

The scope covers the following :

- This NOS covers the development of interactive and scalable web applications using modern front-end frameworks and best practices. It includes building user interfaces with ReactJS, leveraging Next.js for optimized rendering strategies, and implementing unit testing methodologies to ensure application reliability. The modules focus on component-based development, state management, routing, data fetching, server-side capabilities, and testing strategies, providing a comprehensive skill set for full-stack JavaScript development.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** • Demonstrate a solid understanding of React core concepts
• (components, state management, JSX, Virtual DOM).
- PC2.** • Build dynamic and responsive user interfaces (UIs) using React
• components with effective state management techniques (Redux,
• Zustand, or alternatives).
- PC3.** Effectively handle user interactions, data fetching from APIs, and navigation between views within React applications.
- PC4.** Explain the advantages of Next.js over vanilla React and its overall approach to building web applications.
- PC5.** • Structure Next.js projects effectively with pages, layouts, and
• implement routing for smooth user navigation.
- PC6.** • Implement different rendering approaches (SSG, SSR, CSR) within
• Next.js applications, understanding their use cases and performance
• implications.
- PC7.** • Utilize client and server components effectively within Next.js projects
• for a modular and performant architecture.
- PC8.** • Practice data fetching patterns, caching, revalidation techniques, and
• server actions within Next.js applications to optimize performance and maintainability

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	10	10	-	-
PC1. - Demonstrate a solid understanding of React core concepts (components, state management, JSX, Virtual DOM).	1	1	-	-
PC2. - Build dynamic and responsive user interfaces (UIs) using React - components with effective state management techniques (Redux, Zustand, or alternatives).	1	-	-	-
PC3. Effectively handle user interactions, data fetching from APIs, and navigation between views within React applications.	1	1	-	-
PC4. Explain the advantages of Next.js over vanilla React and its overall approach to building web applications.	1	1	-	-
PC5. - Structure Next.js projects effectively with pages, layouts, and - implement routing for smooth user navigation.	1	1	-	-
PC6. - Implement different rendering approaches (SSG, SSR, CSR) within - Next.js applications, understanding their use cases and performance - implications.	1	1	-	-
PC7. - Utilize client and server components effectively within Next.js projects - for a modular and performant architecture.	2	2	-	-
PC8. - Practice data fetching patterns, caching, revalidation techniques, and - server actions within Next.js applications to optimize performance and maintainability	2	3	-	-
NOS Total	10	10	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	ASP/ITS/N0503
NOS Name	Building Web applications with React and Next.js
Sector	Information Technology Sector
Sub-Sector	
Occupation	MERN Stack Developer
NSQF Level	5
Credits	3
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Qualification Pack

ASP/ITS/N0504: Building with Database: MongoDB for developers

Description

This NOS provides a comprehensive understanding of front-end web development using ReactJS and Next.js, along with unit testing practices for robust software development. Learners will gain expertise in building dynamic and reusable UI components, implementing state management solutions like Redux/Zustand, and integrating client-side routing with React Router. The training includes setting up and structuring Next.js applications, exploring different rendering techniques (SSG, SSR, CSR), and optimizing performance using caching and revalidation strategies. Additionally, the NOS covers unit testing with Jest, enabling students to write and execute effective test cases, leverage mock objects, and adopt test-driven development (TDD) principles. Through practical implementation, students will develop scalable, high-performance web applications that adhere to industry best practices.

Scope

The scope covers the following :

- This NOS focuses on equipping learners with the skills to build, optimize, and test modern web applications using ReactJS, Next.js, and unit testing methodologies. It covers interactive UI development, state management, client-side and server-side rendering, data fetching strategies, and testing best practices. The modules provide hands-on experience in developing scalable applications, improving performance, and ensuring code reliability through structured unit testing.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** • Explain the concept of databases and their importance in web applications for data storage and retrieval.
- PC2.** Differentiate between relational and NoSQL databases, understanding the core principles of MySQL and MongoDB.
- PC3.** Identify different MySQL storage engines and when to choose a specific engine for optimal performance (basic understanding).
- PC4.** Demonstrate proficiency in creating and managing MySQL databases and tables using SQL commands.
- PC5.** Write basic SQL queries to insert, update, delete, and retrieve data from MySQL tables.
- PC6.** Utilize the WHERE clause within SELECT statements to filter data based on specific conditions.
- PC7.** Explain the core concepts of MongoDB, including its document-oriented structure and key differences from relational databases.
- PC8.** Create and manage MongoDB databases and collections using tools like the command line or MongoDB Compass.v
- PC9.** Write basic queries using the find command and manipulate results using projection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	10	10	-	-
PC1. • Explain the concept of databases and their importance in web • applications for data storage and retrieval.	1	1	-	-
PC2. Differentiate between relational and NoSQL databases, understanding the core principles of MySQL and MongoDB.	1	1	-	-
PC3. Identify different MySQL storage engines and when to choose a specific engine for optimal performance (basic understanding).	1	1	-	-
PC4. Demonstrate proficiency in creating and managing MySQL databases and tables using SQL commands.	1	1	-	-
PC5. Write basic SQL queries to insert, update, delete, and retrieve data from MySQL tables.	1	1	-	-
PC6. Utilize the WHERE clause within SELECT statements to filter data based on specific conditions.	1	1	-	-
PC7. Explain the core concepts of MongoDB, including its document-oriented structure and key differences from relational databases.	1	2	-	-
PC8. Create and manage MongoDB databases and collections using tools like the command line or MongoDB Compass.v	1	1	-	-
PC9. Write basic queries using the find command and manipulate results using projection	2	1	-	-
NOS Total	10	10	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	ASP/ITS/N0504
NOS Name	Building with Database: MongoDB for developers
Sector	Information Technology Sector
Sub-Sector	
Occupation	MERN Stack Developer
NSQF Level	5
Credits	2
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Qualification Pack

ASP/ITS/N0505: NodeJS for Backend development

Description

This NOS provides a comprehensive understanding of database concepts, relational database management using MySQL, and NoSQL database management with MongoDB. Learners will gain expertise in designing normalized relational databases, writing complex SQL queries with JOINS and aggregate functions, and managing database security. Additionally, they will explore NoSQL data modeling principles, perform CRUD operations in MongoDB, and utilize advanced querying techniques, including aggregation pipelines. The training also emphasizes best practices for database performance, user access control, and security configurations. Through theoretical knowledge and practical implementation, learners will develop the skills necessary for effective data storage, retrieval, and management in modern applications.

Scope

The scope covers the following :

- This NOS focuses on equipping learners with essential database management skills, covering both relational (MySQL) and NoSQL (MongoDB) databases. It includes database design, SQL query writing, data manipulation, security implementation, and optimization techniques. The modules provide hands-on experience in CRUD operations, schema modeling, data analysis, and database administration for building scalable, secure, and efficient database solutions.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** • Demonstrate a solid understanding of Node.js fundamentals
 - (event-driven architecture, non-blocking I/O) and its advantages for
 - web development.
- PC2.** • Utilize modules effectively, manage packages with npm, and structure projects with clear organization (package.json).
- PC3.** • Build dynamic web applications using Express.js, implementing
 - middleware, routing logic, handling user interactions (forms, queries),
 - and connecting to databases.
- PC4.** • Integrate MongoDB with Node.js applications using Mongoose for
 - Object Data Modeling (ODM) and perform CRUD operations (Create,
 - Read, Update, Delete) effectively
- PC5.** • Demonstrate a solid understanding of Node.js fundamentals
 - (event-driven architecture, non-blocking I/O) and its advantages for
 - web development.
- PC6.** • Utilize modules effectively, manage packages with npm, and structure projects with clear organization (package.json).
- PC7.** • Describe various deployment options for Node.js applications (e.g.,
 - Heroku, AWS) and strategies for scaling to handle increased workloads
 - (horizontal & vertical scaling, load balancing).
- PC8.** • Identify and troubleshoot performance bottlenecks within Node.js
 - applications using profiling tools.

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- PC9.**
- Implement caching mechanisms (e.g., Redis) to improve application performance and user experience.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	10	10	-	-
PC1. • Demonstrate a solid understanding of Node.js fundamentals • (event-driven architecture, non-blocking I/O) and its advantages for • web development.	2	1	-	-
PC2. • Utilize modules effectively, manage packages with npm, and structure • projects with clear organization (package.json).	1	2	-	-
PC3. • Build dynamic web applications using Express.js, implementing • middleware, routing logic, handling user interactions (forms, queries), • and connecting to databases.	1	2	-	-
PC4. • Integrate MongoDB with Node.js applications using Mongoose for • Object Data Modeling (ODM) and perform CRUD operations (Create, • Read, Update, Delete) effectively	-	1	-	-
PC5. • Demonstrate a solid understanding of Node.js fundamentals • (event-driven architecture, non-blocking I/O) and its advantages for • web development.	1	1	-	-
PC6. • Utilize modules effectively, manage packages with npm, and structure • projects with clear organization (package.json).	1	1	-	-
PC7. • Describe various deployment options for Node.js applications (e.g., • Heroku, AWS) and strategies for scaling to handle increased workloads • (horizontal & vertical scaling, load balancing).	2	1	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC8. - Identify and troubleshoot performance bottlenecks within Node.js - applications using profiling tools.	1	-	-	-
PC9. - Implement caching mechanisms (e.g., Redis) to improve application - performance and user experience.	1	1	-	-
NOS Total	10	10	-	-

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Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	ASP/ITS/N0505
NOS Name	NodeJS for Backend development
Sector	Information Technology Sector
Sub-Sector	
Occupation	MERN Stack Developer
NSQF Level	5
Credits	2
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Qualification Pack

ASP/ITS/N0506: On-the-job training

Description

This NOS gives hands-on experience in imparting the vital knowledge, skills, and competencies required for candidates to accomplish specific tasks within the workplace.

Scope

The scope covers the following :

- Skill Acquisition
- Workplace Readiness
- Competency-Based Learning
- Holistic Development

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

PC1. On the job training

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	-	100	-	-
PC1. On the job training	-	100	-	-
NOS Total	-	100	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	ASP/ITS/N0506
NOS Name	On-the-job training
Sector	Information Technology Sector
Sub-Sector	
Occupation	MERN Stack Developer
NSQF Level	5
Credits	3
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Qualification Pack

ASP/ITS/N0507: PROJECT ON MERNSTACK DEVELOPER COURSE

Description

This project helps students build a task management app using the MERN Stack

Scope

The scope covers the following :

- The project helps students learn how to build a web app .

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

PC1. PROJECT

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	-	-	45	45
PC1. PROJECT	-	-	45	45
NOS Total	-	-	45	45

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Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	ASP/ITS/N0507
NOS Name	PROJECT ON MERNSTACK DEVELOPER COURSE
Sector	Information Technology Sector
Sub-Sector	
Occupation	MERN Stack Developer
NSQF Level	5
Credits	2
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Qualification Pack

DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

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- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e- mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

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PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings

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- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

Assessment Criteria

Assessment Components & Weightage

ASAP Kerala employs a structured assessment framework that includes Attendance, Assignments, Internal Assessments, and Final Assessments. Assessments align with the National Skills Qualifications Framework (NSQF) and follow a Continuous and Comprehensive Evaluation (CCE) approach.

Assessment Type Max Marks

Assignment 1 35

Internal Assessment 1 50

Assignment 2 35

Internal Assessment 2 50

Qualification Pack

Final Assessment 100

Attendance 30

Total 300

Attendance Criteria

Attendance is crucial, as skill-based sessions involve interactive learning. Students with less than 70% attendance are ineligible for final assessment and certification.

Attendance % Marks

Above 90% 10

86-90% 9

80-85% 8

70-80% 7

Below 70% 0

Internal/Continuous Assessments & Assignments

Assignments (35 marks each): Two assignments, after 25% and 75% of syllabus completion. Evaluated and recorded by trainers.

Internal Assessments (50 marks each): Conducted after 50% and 100% of syllabus completion. Questions include 60% objective and 40% descriptive formats.

Final/End Assessment

Conducted by the Assessment Division of ASAP Kerala, covering:

Practical/Skill Test (at OJT/Skill Development Centers)

Project Presentation & Viva Voce

MCQ & Descriptive Written Exam

Qualification Pack

Assessment Governance & Quality Assurance

Assessment Division of ASAP Kerala oversees the process.

Question Bank prepared by Subject Matter Experts; 20% of questions refreshed annually.

Assessments held at ASAP Skill Parks, Accredited Centers, Colleges, or AI-Proctored Platforms.

ID Verification & AI-based Face Recognition for authentication.

Final Score & Certification issued post evaluation by expert assessors.

Final Score & Grading

Score (%) Grade Description

90-100 A+ Excellent

80-89 A Very Good

70-79 B Good

60-69 C Above Average

50-59 P Pass

Below 50 F Fail

Minimum Aggregate Passing % at QP Level : 50

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

Qualification Pack

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
ASP/ITS/N0501.Introduction to MERN	10	10	0	0	20	10
ASP/ITS/N0502.Javascript and Typescript Fundamentals	10	10	0	0	20	10
ASP/ITS/N0503.Building Web applications with React and Next.js	10	10	0	0	20	10
ASP/ITS/N0504.Building with Database: MongoDB for developers	10	10	0	0	20	10
ASP/ITS/N0505.NodeJS for Backend development	10	10	0	0	20	10
ASP/ITS/N0506.On-the-job training	0	100	0	0	100	30
ASP/ITS/N0507.PROJECT ON MERNSTACK DEVELOPER COURSE	0	0	45	45	90	20
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	0
Total	70	180	45	45	340	100

Logo
Not
Available



Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

Qualification Pack

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.